

Q1. APRIORI ALGORITHM

PseudoCode:

START

Input transaction dataset

Set Minimum Support

Find all individual items

Count Support of each item

Remove items below minimum Support

Keep the remaining items as frequent itemsets

WHILE frequent itemsets are available

 Generate Candidate itemsets

 Count Support of each Candidate

 Remove candidates below minimum Support

 Keep the remaining candidates as frequent itemsets

END WHILE

Display all frequent itemsets

STOP.

Candidate Generation

From the dataset :

1-itemsets :

{Mobile}

{Earphones}

{Charger}

Generate;

2-itemsets;

{ Mobile , Earphones }

{ Mobile , Charger }

{ Earphones , Charger }

If the support is sufficient, generate;

3-itemsets;

{ Mobile , Earphones , Charger }

Q2. FP-Growth Algorithm

START

Input borrowing transactions

Set minimum support

Count the frequency of each book

Remove books below minimum support

Sort remaining books by frequency

Create the FP-Tree

Create the Header Table

For each book in the Header Table

Find its conditional pattern base

Create a conditional FP-Tree

Generate frequent itemsets

END For

Display all frequent itemsets .

STOP.

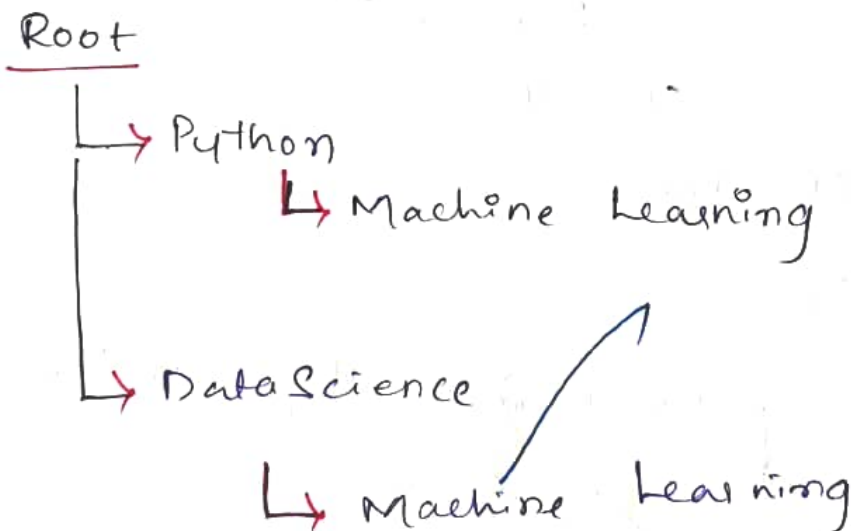
FP - Tree Construction

First count the books :

python → 4
Machine Learning → 4
Data Science → 3

Then arrange the books according to their frequency and insert each Transaction into the FP-Tree

For example :



Q3. Association Rule Generation

Pseudocode:

START

Input frequent itemsets

Set Minimum Confidence

FOR each frequent itemset

Generate possible rules

For each possible rules

Calculate Confidence

IF confidence $\geq$ minimum confidence

Keep the rule

ELSE

Remove the rule

END IF

END FOR

END FOR

Display valid association rules

STOP.



Rule Generation

From:

{Internet, Streaming}

possible rules:

Internet $\rightarrow$ Streaming

Streaming $\rightarrow$ Internet

From: {Streaming, Cloud Storage}

Possible rules:

{Streaming $\rightarrow$ Cloud Storage

Cloud Storage $\rightarrow$ Streaming

Confidence:

confidence tells us how often the second service is purchased when the first service is purchased

Example:

$$\text{Confidence (Internet} \rightarrow \text{Streaming)} \\ = \frac{\text{Support (Internet, Streaming)}}{\text{Support (Internet)}}$$

$$= \frac{3}{4}$$

$$\approx 75\%$$

If the minimum confidence is 70%, thus this rule is accepted.

Q4. Constraint - Based Frequent Itemset Mining

Dataset: Student Course Enrollment

Pseudocode:

START

Input Student course transactions

Set constraint = "Python"

Find all individual courses

Generate candidate itemsets

For each candidate itemset

IF itemset contains "Python"

Calculate Support

IF Support $\geq$ minimum Support

Keep the itemset

ELSE

Remove the itemset

END IF

ELSE

Remove the itemset

END FOR

Generate larger itemsets containing "Python"

Repeat until no more itemsets can be generated,

Display frequent itemsets containing "Python"

STOP.

Candidate Generation ;

from the dataset :

$\{ \text{Python} \}$
 $\{ \text{Python}, \text{Database} \}$
 $\{ \text{Python}, \text{AI} \}$
 $\{ \text{Python}, \text{Database}, \text{AI} \}$

only itemsets containing Python are considered.

Q5. Multilevel Association Rule mining

Dataset : Hospital Treatment Hierarchy

Pseudocode :

START

Input treatment transactions

Input treatment hierarchy

Set minimum Support and Confidence.

Find frequent treatments at the top level

Generate frequent itemsets

For each hierarchy level

Generate frequent itemsets at that level

Calculate Support

Remove itemsets below minimum Support.

Generate association rules

Calculate confidence

Keep rules above minimum confidence

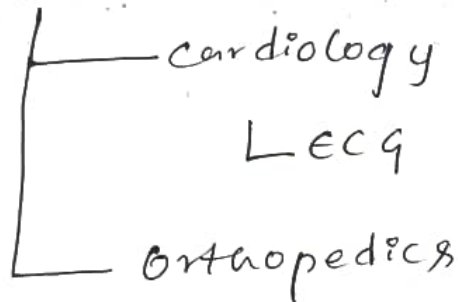
END For

Display rules from all hierarchy levels

STOP.

Treatment Hierarchy

Healthcare



Example:

Higher level:

Healthcare $\rightarrow$ Cardiology

Lower level:

Cardiology $\rightarrow$ ECG

This allows us to discover patterns at both general and detailed levels.

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